Bridging local and technocratic epistemologies for better non-economic loss & damage financing modalities in the Pacific

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Abstract

The Pacific Islands face disproportionate and escalating risks from climate change, manifesting in both economic and non-economic loss and damage (L&D). Non-economic loss and damage (NELD) - encompassing social, cultural, and environmental harms - remains inadequately addressed in existing global mechanisms such as the Warsaw International Mechanism and the Bridgetown Initiative, which often lack epistemologically appropriate, evidence-based policymaking foundations. While existing scholarship has explored NELD typologies, governance arrangements, and measurement approaches, there is a critical gap in tools that translate local epistemologies into technocratic decision-making processes guiding the delivery (modalities) of NELD financing. This paper proposes a novel analytical framework for bridging this epistemological gap - integrating insights from policy translation and anticipatory governance literature to better inform modalities for NELD financing. The framework seeks to enable the incorporation of Pacific Islander knowledge systems into financing modalities for NELD, with the aim of fostering long-term, systemic change within complex socio-political and environmental contexts. While applicable to the whole Pacific region, the paper focuses on Vanuatu as a jurisdiction where this policy translation framework could be applicable.

Keywords: Loss & Damage, Finance, Climate Change, Policy Translation, Epistemology

1. Introduction

The Pacific Islands face numerous catastrophic risks due to climate change, the impacts of which are considered 'loss and damage' (L&D). Loss & damage refers to the unavoidable and irreversible effects of climate change, despite best efforts to mitigate and adapt [1]. Across the Pacific, threats such as Sea Level Rise (SLR), marine heatwaves, cyclones and extreme rainfall events manifest the characteristics of L&D [2].

Figure 1 highlights a typology for non-economic loss & damage (NELD) in the Pacific [3].



Fig. 1. Typologies of NELD in the Pacific [3]

The Pacific has limited capacity to adapt to climate change, despite being a significantly small player in its contribution. This not only indicates climate injustice, but also demonstrates how Pacific islands are susceptible to loss & damage, across economic and non-economic characteristics. In light of this, initiatives such as the Warsaw International Mechanism and the Bridgetown Initiative are exploring financial avenues to address loss & damage, and across social, cultural, economic and governance domains. However, these financial avenues must be informed by epistemologically appropriate evidence-based policymaking [4]. Some critics have argued that these initiatives have not been based on such evidence [5-6]. In response, Parties at COP27 (2022) agreed to establish a dedicated Fund for Responding to Loss and Damage (FRLD), which was formally operationalised at COP28 [7]. This new fund marks a milestone in principle; however, by late 2024, it had only amassed approximately USD 700 million in initial pledges, a minuscule fraction of the hundreds of billions needed annually. Ensuring the FRLD effectively addresses NELD (not just infrastructure losses) remains a significant challenge, highlighting the persistent gap between high-level ambitions and local realities [7-8].

While there is literature exploring potential governance and metrics to measure NELD and its financing [8], there is a significant gap in literature assessing how to actually best deliver NELD

financing through local modalities for change. This is with the intent of driving long-term systemic change in a complex operating environment.

Therefore, this paper proposes a new way of analysing the complex translation of local epistemologies into the modalities of NELD financing, drawing upon existing policy translation and anticipatory governance literature [9-11].

While applicable to the whole Pacific region, the paper focuses on Vanuatu as a jurisdiction where this policy translation framework could be applicable.

2. Scope and Objectives

The paper will prospect the possibility of a new analytical framework for assessing the translation of local epistemologies into financial allocations for non-economic loss & damage policy delivery. It will go beyond some existing literature, which looks at the translation of local epistemologies into policy decision-making, and instead also assesses avenues for the actual modalities (i.e. delivery methods) of NELD financing.

The intention is to provide a new option for how communities and governments in the Pacific might begin to reconsider and reframe the translation of local epistemologies into the tangible modalities of delivery for NELD financing. The paper does not claim this is the only or necessarily best way to translate local epistemologies into NELD financing modalities, but it offers a framework for consideration, which is open to further critique and empirical assessment.

This involves:

- Assessing the literature of existing gaps in NELD financing in the Pacific, both in the design of policies and, importantly, in the modalities of delivery (both grey and peer-reviewed literature). It will focus on Vanuatu as a case study.
- Assessing how existing policy translation and anticipatory governance literature might support Pacific island communities and governments to better understand the impactful modalities of NELD financing in the face of epistemological contentions and uncertainties, such as the policy taxonomy of Cashore & Howlett [10] and the ‘pathways of influence’ [11].

This paper has several limitations. Namely, it is an exploratory document for further research, meaning more qualitative and empirical critique of the proposed framework is necessary when exploring the more nuanced application of localised epistemologies to the delivery of NELD financing.

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3. Setting the Scene: Existing Literature and Knowledge Gaps

The Pacific Islands face some of the most severe and irreversible impacts of climate change. While scholarship acknowledges the importance of non-economic loss and damage - such as cultural heritage, identity, and environmental degradation - most literature remains concentrated on *defining* or *designing* policy, with insufficient attention to the practical delivery (i.e. modalities) of financing in ways that reflect and respect local, indigenous epistemologies [12-14].

3.1 Institutional Inadequacies: Implementing Top-Down approaches, Not Bottom-up

Many existing funding NELD mechanisms, such as conditional loans, leave communities vulnerable to climate change for the long-term, often only solving issues with a short-term, non-durable focus [14-16].

Conversations around financing NELD in the Pacific often focus on high-level constructs, typologies, and mechanisms, while glossing over how these are actually delivered on the ground [14]. For instance, case-based critiques underscore that donor-driven financing often privileges short-term, externally focused outcomes - such as quick infrastructure projects - at the expense of long-term community resilience, resulting in fragmented implementation and persistent bottlenecks [17]. Likewise, institutional analyses further highlight the high compliance demands and rigid operational frameworks imposed by donors, which pull governmental focus toward accountability upward, rather than empowering communities through meaningful delivery modalities [18]. The new Fund for Responding to Loss and Damage (FRLD) hosted by the World Bank, has drawn scrutiny over how a top-down institution will ensure transparent, climate justice-oriented disbursement.During negotiations, many developing countries opposed placing the fund at the World Bank due to concerns about its governance and track record, preferring an independent fund – a proposal ultimately overruled by developed countries' push for the Bank as host [18]. Critics note that the UNFCCC's existing system often reinforces a one-size-fits-all approach, struggling to channel locally specific, culturally diverse data into global funding decisions. Without reforms to prioritize bottom-up input and community voices, even well-intended mechanisms risk reproducing the misalignments that hinder effective support for non-economic losses and damages [7]. Put simply, there is a tension between global frameworks and local realities: Pacific experts have observed that the institutionalized UNFCCC narrative can be at odds with the local priorities and realities of vulnerable communities, which raises doubts about how well the FRLD can meet diverse on-the-ground needs under its current structure [5-7].

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3.2 A Disconnect Between Planning and Ground Reality

Policy primers and roadmaps - such as those exploring regional climate finance opportunities - elaborate financing pathways but often remain abstract. There is limited evidence or documentation of local epistemologies being used to shape delivery approaches themselves [18-19]. Without this grounding, financial tools risk perpetuating top-down, standardised solutions that are misaligned with community priorities. One clear manifestation of this disconnect is in climate-related mobility and relocation. When funding focuses narrowly on physical infrastructure or resettlement logistics, it can inadvertently create new losses, such as families uprooted from ancestral land, kinship networks fragmented, social traditions lost, and communal ways of life irreversibly altered in resettlement [20]. Pacific communities often resist externally driven relocation plans due to strong cultural attachments (e.g. the ties to vanua that can lead to voluntary immobility), knowing that a forced move may trigger cascading non-economic losses – from weakened communal support and loss of spiritual connection to place, to an erosion of identity and "ontological security" that is nearly impossible to compensate [20-21]. These outcomes highlight why planned relocations must be guided by local epistemologies and values. Fiji's experience provides a telling example: early relocations in Fiji revealed severe social disruption, prompting the development of Planned Relocation Guidelines and Standard Operating Procedures (SOPs) to ensure cultural continuity and community agency during climate-induced moves [22-23]. By coupling relocation projects with a dedicated national trust fund and community consultations, Fiji's framework aims to address physical safety without inflicting further NELD – an approach other Pacific nations could emulate [22-23].

3.3 Emerging Recognition, Yet Sparse Practice

Although recent work emphasises the diversity and existential importance of Pacific perspectives on NELD - highlighting cultural loss, sense of place, and deep identity attachments [20], there is still a significant theory–practice gap. The literature does not yet offer *operational frameworks or tools* to guide how local epistemologies might shape actual financial instruments, channels, or delivery mechanisms [12].

For example, in Fiji, the Indigenous concept of vanua encapsulates the inseparable connection between land, community, and identity; consequently, the loss of ancestral land or relocation of a village represents a profound spiritual and cultural rupture beyond any economic valuation [23-25]. Likewise, well-being in Indigenous Fijian communities is grounded in values such as veilomani (kindness and communal love), veikauwaitaki (caring for one another), and bula vakayalo (spiritual life), underscoring that NELD encompasses relational and spiritual dimensions often overlooked by technocratic frameworks [25]. One key gap is the ability to measure and quantify these intangible losses in culturally appropriate ways. Conventional climate finance metrics – geared toward monetary damage – cannot capture phenomena like the loss of oral histories, breakdown of social cohesion, or diminished spiritual fulfillment. Pacific

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scholars are therefore calling for value-based frameworks that embed Indigenous principles of reciprocity, communal solidarity, and spirituality into indicators for loss and damage [25-26]. Emerging initiatives hint at how this might be done: Fiji's forthcoming Comprehensive Risk and Vulnerability Assessment Methodology (CRVAM), for instance, is being designed to integrate NELD considerations (e.g. the loss of cultural sites, traditional knowledge, or mental well-being) into community risk assessments [27-29]. By developing such context-specific indicators, Pacific stakeholders hope to make 'intangible' losses visible within formal finance mechanisms, enabling more responsive and just allocation of funds [28].

4. The Policy Translation Framework: Exploring a New Methodology for Translation Epistemologies into Policy

The paper will explore the application of the following methodological frameworks, when applied together, as a way of prospecting the translation of localised epistemologies into financial allocations and delivery for loss & damage funds to achieve systemic, long-term impact.

The following methodological frameworks are 'gateways' to sequentially help translate technocratic financial decision-making into modalities for delivery.

1. **Narrative, visuals and data [9], to;**
2. **Policy taxonomy (policy ends and means) [10], to;**
3. **Pathways of Influence [11].**

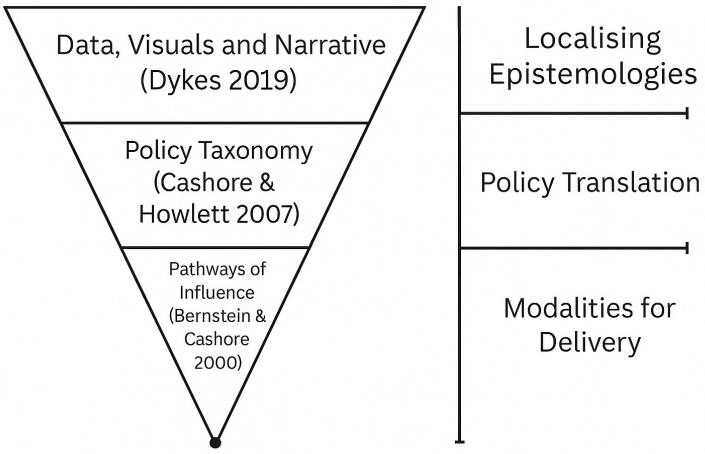


Fig. 2. Proposed policy translation 'gateways' for the translation of local epistemologies into NELD financing modalities.

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4.1 Gateway 1: Data, Visuals and Narrative - Localising Epistemologies

Data storytelling - which combines narrative, data, and visuals - is a powerful way to turn insights into action, making information more memorable, persuasive, and engaging than raw statistics alone. Drawing on Dykes [9], there are 3 key components to creating storytelling through data and decision-making.

1. **Narrative**: Following classic storytelling arcs (e.g. Hero's Journey), a data story begins by setting the scene, leads to a central insight, and ends with actionable recommendations.

2. **Data**: Effective data storytelling involves selecting the most relevant data, simplifying complexity by aggregating or removing secondary details, and focusing on clarity over comprehensiveness.

3. **Visuals**: Good visuals use the appropriate chart type for the message, calibrate layout and design for audience goals, and use techniques like colour sparing, text emphasis, and staged reveals to highlight key insights and reduce clutter.



Fig. 3. Narrative, visuals and data framework [9]

The data storytelling framework of Dykes [9] - integrating narrative, data, and visuals - offers a valuable approach for translating local epistemologies (i.e., place-based knowledge, oral histories, and lived experiences) into forms that are legible, compelling, and actionable for Loss and Damage (L&D) financing mechanisms in the Pacific. For example:

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1. **Narrative**: Pacific communities have rich storytelling traditions rooted in oral histories, cultural memory, and spiritual connections to land [3]. Framing L&D claims through locally grounded narratives - such as stories of displacement, cultural loss, or ancestral ties - can humanise impacts and align with global discourses of justice and dignity.

2. **Data**: Local knowledge often includes qualitative observations, seasonal patterns, and experiential data that may be undervalued in technical assessments [3]. Applying data storytelling principles allows these insights to be structured, prioritised, and framed in ways that resonate with international funding institutions without erasing their original meaning.

3. **Visuals**: Traditional knowledge can be powerfully communicated through maps, oral-visual formats, and symbolic representations (e.g. carvings, drawings, time-lapse imagery). These visuals, if respectfully co-produced, can bridge the communication gap between community experience and funder expectations [4].

In short, storytelling frameworks like Dykes' offer a means of translating Pacific ways of knowing into policy-relevant formats, enabling more just and effective access to L&D finance while preserving cultural integrity and epistemological sovereignty.

4.2 Gateway 2: Policy Taxonomy - Policy Translation

Cashore & Howlett [10] argue policy must be packaged in a coherent manner, disaggregating ways of analysing policy:

- **Policy ends** - this is what policy aims to achieve, referring to the substantive goals or outcomes a policy seeks to produce. It is analysed into the following components:

Goals (Abstract ends): Broad, often normative or ideological statements (e.g., "promote climate justice" or "reduce climate vulnerability").

Objectives (Operationalised ends): More specific and measurable aims that interpret the goals in policy-relevant terms (e.g., "relocate 5,000 residents from high-risk coastal areas by 2035").

Settings (Calibrated ends): The quantitative or prescriptive thresholds that detail how the objectives are to be achieved (e.g., "allocate $10 million annually to relocation programs").

- **Policy means** - how policy is implemented. These are the instruments and strategies used to pursue the policy ends.

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Instrument types (Abstract means): Broad categories of tools, such as regulatory, economic, informational, or voluntary approaches.

Instrument logic/design (Instrumental objectives): The choice and design rationale for specific tools within the category (e.g., why use direct grants over concessional loans).

Instrument settings (Calibrated means): The specific technical parameters of an instrument (e.g., "grant covers 80% of household relocation costs, up to $25,000").

This is presented through the following taxonomy:

		High level abstraction (policy orientation)	Operationalization (program)	On the ground specification (measures)
Policy Content	Policy ends (aims)	**GOALS** **What general types of ideas govern policy development?** e.g. environmental protection, economic development, Social cohesion	**OBJECTIVES** **What does policy formally aim to address?** e.g. saving wilderness or species habitat, reducing greenhouse gas emissions	**SETTINGS** **What are the specific "on the ground" requirements of the policy?** e.g. size of protected areas, level of carbon tax
	Policy means (instruments)	**INTERVENTION LOGIC** **What general norms guide policy instrument preferences?** e.g. coercive "command and control", voluntary, markets, neoliberal norms	**TOOLS** **What types of instruments are utilized?** e.g. tax incentives, loans, public enterprise, cap and trade carbon markets	**CALIBRATIONS** **What are the specific ways in which the instrument is applied?** e.g. qualification for tax incentives, rules governing cap and trade markets such as specifics on leakage, allocation of resources and approach to enforcement

Fig. 4. Policy taxonomy framework, showing ends and means, alongside scales of policy design [10]

The disaggregation of policy taxonomy - particularly the expansion of three orders of change into a more granular six-level framework - offers a powerful lens for understanding the complex, layered dynamics of Loss and Damage (L&D) and climate-related policy change in the Pacific Islands. By parsing out the different elements of a policy regime - goals, objectives, and setting (**policy ends**), and intervention logic, tools and calibrations (**policy means)** - Pacific policymakers, analysts, and advocates can better track, influence, and design effective interventions that are context-appropriate and adaptable to local and global shifts.

Applying the Cashore & Howlett [10] *ends–means* taxonomy to Pacific L&D policy surfaces not only hidden misalignments and implementation gaps, but also shows *where* and *how* data, narrative, and visuals should be deliberately deployed to translate ambition into culturally

grounded, fundable, and trackable action. Specifically, this is when integrated with data, visuals and narrative, as shown in Table 1 and Table 2.

Table 1 - Achieving *Clarity in Policy Ends* For NELD (What we are trying to accomplish?)

Policy Layer (Ends)	Typical Pacific L&D Challenge	Role of Data	Role of Narrative	Role of Visuals	Integrated Payoff
Goals (Abstract)	Broad aspirational phrases ("enhance resilience", "protect cultural identity") are normatively strong but decision-blunt.	Macro indicators (exposure, cultural/non-economic loss typologies) benchmark why the goal matters.	Cultural and ancestral place-based stories embed moral salience & justice framing.	High-level conceptual schematics (e.g., "resilience ecosystem" diagrams) crystallize paradigm.	Converts diffuse ideals into a shared paradigm funders & communities both endorse.
Objectives (Operational)	Goals under-specified (who, what scale, by when).	Disaggregated risk, displacement projections, livelihood impact baselines quantify scope.	Community voice lends legitimacy to prioritisation (why *these* villages first).	Story maps / timelines align projected impacts with planned phases.	Prioritisation matrix that is technically robust and socially legitimate.
Settings (Calibrated)	Missing or donor-driven numeric targets weaken accountability.	Precise metrics (e.g., # households relocated, % customary land secured) enable monitoring.	Micro-narratives explain trade-offs behind chosen thresholds (why 3 villages/year).	Dashboard visuals & before/after coastal change imagery make progress tangible.	Creates a transparent results framework resistant to goal drift.

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Table 2 - Achieving *Clarity in Policy Means* For NELD (How we are trying to accomplish our objectives?)

Policy Layer (Means)	Typical Pacific L&D Challenge	Role of Data	Role of Narrative	Role of Visuals	Integrated Payoff
Instrument Types (Abstract Means)	Over-reliance on externally favoured tools (insurance, one-off grants) vs. locally fit blends.	Comparative evidence on cost-effectiveness, risk layers, and equity impacts.	Framing instruments through lived experience (e.g., why grants > loans for customary relocation) shifts preference legitimacy.	Comparative tool typology infographics clarify functional differences.	Justifies a context-tailored policy mix rather than imported templates.
Instrument Logic / Design (Operational Means)	Rationale for tool combination not explicit; sequencing opaque.	Scenario & sensitivity analyses show why staged relocation fund + contingency facility reduces cumulative loss.	Process narratives capture co-design steps, reinforcing procedural justice.	Flowcharts & causal diagrams show sequencing (trigger → payout → relocation phase).	Strengthens internal coherence and donor confidence in design logic.
Instrument Settings (Calibrated Means)	Vague eligibility, payout formulas, and fiduciary thresholds stall implementation.	Micro-level cost data, damage classification, and funding gap analyses refine parameters.	Beneficiary stories surface edge cases guiding inclusive criteria.	Parameter tables, choropleth risk maps, and annotated payout ladders aid rapid uptake.	Delivers actionable, auditable implementation

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Finally, the article will explore ‘pathways of influence’ [11] that help drive transformative, long-term change by shifting the actual modalities of delivery of NELD financing.

4.3 Gateway 3: Pathways of Influence - Modalities for Delivery

Bernstein & Cashore [11] propose ‘pathways of influence’ that drive long-term, durable change, cutting through intervening variables that attempt to de-rail long-term systemic change (see Figure 5).

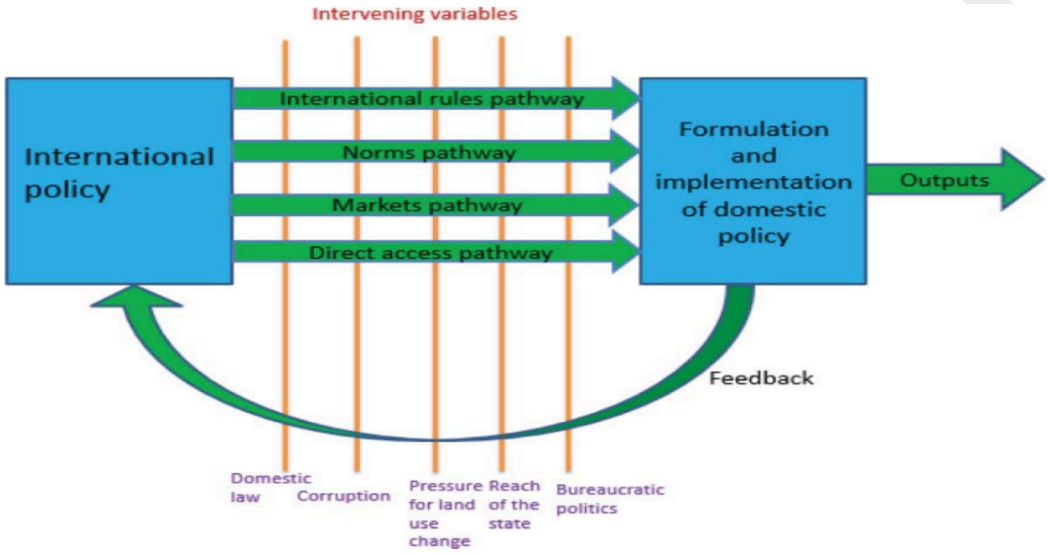


Fig. 5. Pathways of Influence [10] (Image: Humphreys et al. 2017).

Specifically, this involves:

1. **Rules pathway** – Influence occurs through the creation or modification of formal and informal rules, agreements, and standards that guide policy behaviour.

2. **Norms pathway** – Influence arises by shaping shared social values, expectations, and ideas about what is appropriate or desirable.

3. **Markets pathway** – Influence is exerted through economic incentives, disincentives, and the creation or shaping of markets.

4. **Direct access pathway** – Influence is achieved by empowering actors directly through capacity building, funding, and technology transfer.

When integrated with Dykes [9], and the policy ends of Cashore & Howlett [11], it looks like Table 3.

Table 3 - Integration of 'Pathways of Influence' into the other Gateways of Policy Translation (Policy Ends)

Policy Ends	Role of Visuals, Data and Narrative	Related Pathways of Influence	Explanation
Goals (Abstract)	- **Data**: Macro indicators (e.g. cultural loss typologies) justify why the goal matters. - **Narrative**: Cultural and ancestral stories make goals morally compelling. - **Visuals**: Conceptual schematics (e.g. "resilience ecosystems") communicate complexity intuitively.	Norms + Rules	Abstract goals like "enhance resilience" shape *what is seen as desirable* (Norms), while framing these goals in national or regional strategies gradually institutionalises them (Rules). Data lends legitimacy; narratives mobilise values; visuals build shared conceptual frames.
Objectives (Operational)	- **Data**: Disaggregated risk and displacement baselines define the scope. - **Narrative**: Local voices justify prioritisation (e.g., which communities first). - **Visuals**: Timelines and story maps align future impacts with planned actions.	Rules + Direct Access	Operational objectives clarify *who*, *when*, and *how*. By translating goals into clear mandates, they establish *rules for action*. Co-created objectives also support capacity-building and local empowerment—thus advancing *direct access*.

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Settings (Calibrated)	- **Data**: Specific metrics (e.g., # households relocated) support monitoring. - **Narrative**: Stories explain trade-offs in thresholds (e.g., why 3 villages/year). - **Visuals**: Dashboards and "before/after" imagery track progress.	Rules + Direct Access	Quantitative targets create concrete policy obligations (Rules), while embedding feedback loops that empower local implementers with the information and tools needed for action (Direct Access).

As can be seen in the above table, Norms and Rules pathways dominate at the goal-setting stage, while Direct Access grows in importance as objectives and calibrated settings determine who acts and with what support. Data, narratives, and visuals help translate intangible aspirations into enforceable and inclusive mandates.

Additionally, when Bernstein & Cashore [11] is overlaid with Dykes [9] and the policy means of Cashore & Howlett [10], it looks like Table 4.

Table 4 - Integration of 'Pathways of Influence' into the other Gateways of Policy Translation (Policy Means)

Policy Means	Role of Visuals, Data and Narrative	Related Pathways of Influence	Explanation
Instrument Types (Abstract Means)	- **Data**: Comparative cost-effectiveness and equity metrics inform choices. - **Narrative**: Community framing shifts legitimacy (e.g., why grants not loans). - **Visuals**: Infographics show trade-offs across tools.	Norms + Markets	The choice of tools is shaped by perceptions of *appropriateness* (Norms), and by their economic incentives and disincentives (Markets). Visuals and narrative help reframe imported tools in ways that better fit Pacific contexts.

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Instrument Logic / Design (Operational Means)	- **Data**: Scenario modelling reveals logic behind sequencing tools. - **Narrative**: Process stories validate procedural fairness and co-design. - **Visuals**: Flowcharts clarify implementation phases.	Rules + Direct Access	Making the rationale behind tool combinations explicit creates more coherent policy rules. Co-design processes and narratives support capacity-building and local ownership—hallmarks of the Direct Access pathway.
Instrument Settings (Calibrated Means)	- **Data**: Micro-costs and risk classifications refine eligibility. - **Narrative**: Local stories surface overlooked needs and edge cases. - **Visuals**: Parameter tables and risk maps support field deployment.	Rules + Direct Access + Markets	Precise implementation parameters shape *how money flows, who benefits, and under what conditions* - the core of both *market incentives* and *policy rules*. Direct access is enabled when communities can understand and navigate these criteria.

As can be seen, Rules and Direct Access pathways are most activated when implementing instruments, while Markets and Norms influence which tools are adopted and how they're legitimised. Integration of local data and storytelling ensures instruments are not just efficient, but equitable and politically resonant.

5. Case Study: Exploring Vanuatu as an Applied Example of this Framework

5.1 Overview of NELD in Vanuatu, and the Need for Better Policy Translation into Modalities for NELD Financing Delivery

Vanuatu is among the most climate-vulnerable nations globally, facing existential risks such as sea-level rise, intensified cyclones, and the loss of cultural heritage sites. As a leading advocate for loss and damage (L&D) financing on the global stage, Vanuatu has advanced its policy design through national climate strategies, active participation in the Warsaw International Mechanism, and engagement with the Bridgetown Initiative. However, like many Pacific Island states, it continues to face a persistent gap between well-articulated frameworks and their practical delivery - especially for non-economic loss and damage (NELD), where impacts are intangible, culturally embedded, and often community-specific [30-31].

Vanuatu's Revised and Enhanced NDC (2021–2030) [30] provides a compelling reflection of its ambitions to directly address NELD while also acknowledging the associated delivery

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challenges. Included among its 12 specific Loss & Damage commitments, the NDC emphasises the country's need for international support in the form of finance, technology, and capacity development to effectively operationalise these priorities [30-31]. The document recognises the limits of adaptation and urges funding mechanisms that can capture losses to cultural heritage, traditional knowledge, and community cohesion - the very dimensions often overlooked in conventional climate finance [21]. Moreover, Vanuatu's NDC signals its commitment not only to engage constructively with multilateral mechanisms, but also to establish actionable pathways through its national Loss & Damage Policy and Implementation Roadmap - a document defining locally grounded typologies of NELD and time-bound actions aligned with community priorities [31].

Through these commitments, Vanuatu is bridging the conceptual gap between the recognition of non-economic losses and the establishment of financing modalities attuned to local epistemologies. By embedding NELD explicitly in both pledges and planning instruments, the nation lays the groundwork for more culturally responsive, operationally realistic delivery mechanisms, including in the context of issues like disaster-induced displacement [32]. However, as the implementation roadmap unfolds, the critical test will lie in translating these well-intentioned NDC commitments into community-centered financing structures that honour kastom knowledge and enable timely local access to loss and damage support.

The aftermath of Cyclone Pam in 2015 provides a clear illustration of this gap. While international assistance flowed in, much of the funding prioritised rebuilding physical infrastructure, with comparatively little directed towards restoring disrupted cultural practices, rehabilitating sacred sites, or supporting the continuation of kastom ceremonies - areas identified by affected communities as essential to recovery [33]. Embedding local epistemologies within financing mechanisms would help address such imbalances. This could involve formalising the role of community knowledge systems, oral histories, and customary governance in shaping fund allocation criteria, ensuring that heritage restoration, spiritual site protection, and intergenerational knowledge transmission are treated as legitimate funding priorities [33-35].

Vanuatu's geography - an archipelago of over 80 islands - underscores the need to diversify delivery channels. Reliance on centralised government ministries can delay access for remote communities. Lessons from the Vanuatu Community Resilience Framework and the Vanuatu Climate Finance Forum suggest that involving provincial councils, NGOs, and faith-based organisations can improve responsiveness and tailor delivery to local governance norms [35-36]. This decentralisation proved valuable during COVID-19 recovery programmes, where non-state actors enabled quicker mobilisation of resources to rural islands when national capacity was stretched [34].

Institutional capacity is another critical determinant of effective NELD delivery. Vanuatu's experience in accessing the Green Climate Fund (GCF) - including delays due to compliance and

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reporting requirements - illustrates the need for streamlined processes that match local administrative realities [37-38]. Investing in context-appropriate digital tools, designed for low-connectivity environments, could help monitor NELD outcomes without overburdening community implementers. Integrating anticipatory governance could further strengthen NELD financing. Vanuatu's National Policy on Climate Change and Disaster-Induced Displacement [32] already anticipate community displacement from climate impacts; linking these proactive plans with pre-arranged financing channels would allow funds to flow quickly when triggers are met [39-40]. This would move the system from reactive appeals to pre-negotiated, context-sensitive action.

Finally, continuous feedback loops are essential. An adaptive, learning-by-doing approach - drawing on the Vanuatu Humanitarian Partnership's post-disaster monitoring practices - could evaluate financing effectiveness not just through financial metrics but by tracking cultural continuity, community cohesion, and other social indicators valued by Ni-Vanuatu communities [41].

By integrating local epistemologies into delivery modalities - through community-based governance, decentralised channels, strengthened institutional capacity, anticipatory planning, and adaptive feedback - Vanuatu could transform strong policy design into tangible, culturally resonant NELD outcomes [41]. This would position the country not only as a global advocate for L&D but also as a regional leader in operationalising climate finance delivery that honours Pacific knowledge systems.

Table 5 and Table 6 demonstrate what this looks like for Vanuatu's context.

Table 5 - Applying the Policy Translation 'Gateways' Presented in Figure 2 to NELD in Vanuatu (Policy Ends)

Policy Ends	Role of Visuals, Data and Narrative	Related Pathways of Influence	Integrated payoffs
Goals (Abstract)	Macro indicators (e.g., cultural loss risk maps, heritage exposure indices); oral histories of kastom and Cyclone Pam's cultural impact; "resilience + heritage" conceptual diagrams.	Norms, Rules	Anchors L&D financing to safeguard cultural identity - e.g., heritage restoration receives dedicated funding as part of recovery frameworks. UNESCO's assessment estimating **USD 1.4 million** needed for cultural infrastructure after Cyclone Pam shows how this becomes actionable [33].

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Objectives (Operational)	Disaggregated risk projections for specific islands; community testimonies prioritising heritage sites; story maps aligning projected losses with intervention timelines.	Norms, Direct Access	Enables funding criteria that prioritise kastom site restoration in vulnerable communities - e.g., Ambae intangible heritage documentation via the Heritage Emergency Fund, making oral traditions fundable components of relocation packages [41].
Settings (Calibrated)	Quantified metrics (e.g., number of nakamals rehabilitated); narratives explaining why, say, 5 sacred sites/year; dashboards showing cultural recovery progress.	Rules, Direct Access	Builds enforceable targets into financing mechanisms. For example, the **Vanuatu Community-based Climate Resilience Project (VCCRP)** includes logic frameworks that can be expanded to monitor NELD outcomes tied to cultural continuity [34].

Table 6 - Applying the Policy Translation 'Gateways' Presented in Figure 2 to NELD in Vanuatu (Policy Means)

Policy Means	Role of Visuals, Data and Narrative	Related Pathways of Influence	Integrated payoffs
Instrument Types (Abstract Means)	Comparative analyses showing effectiveness of cultural fund vs. infrastructure grants; narratives of how heritage preservation contributed to resilience; infographics of instrument options.	**Markets, Norms**	Supports creation of niche funding instruments - e.g., a **Heritage Recovery Grant** stream alongside infrastructure rebuilding funds - validating cultural resilience economically and socially (as evidenced by malvatumauri nakamal survival during Pam) [33].
Instrument Logic (Operational Means)	Scenario models where funds tied to heritage	**Rules, Direct Access**	Enables financing tied to community-defined triggers - for instance, releasing funds when

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	preservation reduce social disruption; narratives on co-design with provincial councils and chiefs; flowcharts of trigger-based funding.		traditional ceremonies resume after a cyclone, modeled on the Vanuatu planned relocation guidelines aligning displacement triggers and funding [32].
Instrument Settings (Calibrated Means)	Micro-cost data for site restoration; beneficiary stories about access needs; annotated application forms and maps showing eligible heritage zones.	**Rules, Direct Access, Markets**	Produces locally accessible funding protocol - for example, simplified application procedures for remote communities to restore sacred sites, with preset grants per site, aligning with the decentralized delivery approach seen in the Vanuatu Community Resilience Framework [35-36].

5.2 Implications and Analysis

For NELD financing in Vanuatu to be culturally resonant and effective, financiers should integrate narrative, data, and visuals into funding frameworks. Narratives grounded in Ni-Vanuatu experiences - such as oral histories and community storytelling - can convey the lived realities of non-economic losses, including disrupted *kastom* knowledge and damaged sacred sites, in ways statistics alone cannot [40-41].

Culturally relevant data - such as the number of traditional ceremonies restored or continuity of customary land stewardship - should complement conventional metrics [42]. Vanuatu's *National Policy on Climate Change and Disaster-Induced Displacement* [32] demonstrate how social and cultural indicators can inform adaptation planning.

Visual tools like participatory mapping and infographics can bridge communication between local actors and financiers, illustrating how environmental loss cascades into cultural erosion [39-40]. Together, these approaches can align financing with Ni-Vanuatu governance norms and commitments in its *Revised and Enhanced NDC* [30].

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Applying Bernstein & Cashore's pathways of influence [11] to Vanuatu highlights how culturally grounded methods can build durable policy and financing change in complex governance landscapes. For example:

Shaping problem definitions (Norms Pathway) – Ni-Vanuatu narratives, such as those captured in community consultations during the *Pla* development, reframe NELD beyond economic metrics to include spiritual, genealogical, and *kastom*-based losses. By embedding these stories in financing proposals, problem definitions shift from abstract climate loss to tangible impacts that matter for identity and sovereignty. In Vanuatu, Cyclone Pam revealed that recovery centred on rebuilding infrastructure overlooked cultural and social harms, such as the loss of sacred sites and interruptions to *kastom* practices [40-41]. UNESCO and post-Pam research showed that restoring cultural heritage and social continuity was as critical to recovery as repairing physical assets [33-35]. By embedding these narratives into funding proposals, Ni-Vanuatu actors have reframed the policy problem from "infrastructure loss" to non-economic loss and damage (NELD) that includes identity, heritage, and social cohesion - pushing funders to recognise and finance such priorities [41-42].

Shifting actor incentives and coalitions (Norms Pathway) – Locally relevant data, for example metrics on the continuity of customary land stewardship or revival of disrupted ceremonies, creates common ground between traditional authorities, provincial planners, national ministries, and international donors. This shared evidence base incentivises coalitions that cross governance levels - critical in Vanuatu where customary systems co-exist with formal state structures. Vanuatu's commitments in the *Revised and Enhanced NDC* [30] show how government, provincial councils, and civil society can co-produce culturally relevant indicators (e.g., measures of cultural continuity or relocation needs). The Green Climate Fund's readiness programmes and resilience projects in Vanuatu illustrate how donor interest grows when evidence links cultural outcomes to community stability and co-benefits - though administrative frictions remain [13;41-42].

Altering institutional logics (Direct Access Pathway) – Visual and participatory tools, such as village-level cultural mapping, can be institutionalised into government monitoring systems. This changes the operating logic from donor-driven technical assessments to co-produced, culturally legitimate measures of resilience [36]. Over time, this helps ensure that NELD financing mechanisms are not just externally compliant but locally owned and thus more durable under shifting political and climate conditions [42-43]. Initiatives such as the integration of participatory mapping of cultural sites, oral histories, and social continuity indicators into formal processes [35-36]. This shift enables funding triggers tied to culturally defined loss thresholds, rather than purely numeric infrastructure metrics [38-39]. Post-Pam evaluations and national finance reviews confirm both the necessity of - and technical challenges to - embedding such metrics into institutional practice [35-36].

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6. Conclusion

Across the Pacific Islands, addressing non-economic loss and damage (NELD) arising from climate change presents a profound and complex challenge. The intangible impacts on culture, identity, and community well-being demand approaches that go beyond conventional economic metrics and technocratic policy designs [3]. This paper has underscored the importance of developing modalities that can effectively translate local epistemologies - rooted in indigenous knowledge systems, oral traditions, and communal values - into tangible modalities of financing mechanisms. Such translation is essential not only for ensuring that loss and damage finance is equitable and just, but also for enhancing the cultural legitimacy and operational effectiveness of climate responses in the region [42]. Indeed, as the international community moves to operationalise new funding arrangements like the FRLD, it will be crucial to design these mechanisms in a bottom-up, inclusive manner. Global funds must adopt governance practices that integrate Indigenous knowledge and community-defined priorities; otherwise, top-down institutions risk perpetuating the very epistemological divide that this study highlights [36-38].

Narratives, data, and visuals collectively form an indispensable medium for this translation process. Across Pacific island nations, knowledge about non-economic losses is often embedded in oral histories and symbolic representations, which may not align neatly with conventional forms of quantitative evidence favoured in international climate finance [42]. By harnessing storytelling techniques paired with culturally relevant data and visuals, Pacific communities can make their experiences and needs visible and persuasive to policymakers and funders, bridging the epistemological divide between local realities and formal decision-making structures [43].

Understanding the pathways of influence within the multi-layered governance systems common to Pacific states is equally critical [40]. Knowledge flows through customary authorities, local governments, national agencies, and international actors, shaping policy outcomes at multiple scales. Mapping and strategically engaging with these pathways allows for enhanced community agency and ensures that financing mechanisms are responsive to the diverse voices and priorities within island societies [42-43].

The policy taxonomy framework offers further analytical clarity by categorising policy instruments in terms of goals, implementation style, and delivery mechanisms. For Pacific Island contexts, this framework aids in identifying financing modalities that are best suited to local governance arrangements and cultural contexts, favouring participatory and adaptive approaches over rigid, top-down models. Applying such a taxonomy highlights gaps between policy intentions and delivery realities, providing guidance for refining instruments to better translate local knowledge into actionable financing [43].

Focusing specifically on Vanuatu, these conceptual tools converge to offer promising avenues for improving the delivery of NELD financing. Vanuatu's experience with Cyclone Pam recovery revealed the limitations of predominantly infrastructure-focused finance and the need to better

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incorporate cultural and spiritual dimensions into funding priorities [33-34]. By embedding *kastom* knowledge into fund allocation criteria, diversifying delivery channels through local councils and NGOs, and strengthening institutional capacity adapted to low-connectivity environments, Vanuatu can enhance both the legitimacy and effectiveness of NELD financing. Moreover, anticipatory governance frameworks, such as those linked to *National Policy on Climate Change and Disaster-Induced Displacement* [32], can align funding flows with emerging climate risks proactively. Embedding continuous feedback mechanisms will ensure that financing adapts to community needs over time, fostering resilience that is both culturally and socially grounded.

In sum, the experience of Vanuatu exemplifies how Pacific Island nations can lead in operationalising locally grounded, epistemologically attuned financing models for non-economic loss and damage. By integrating narratives and visuals, leveraging pathways of influence, and applying policy taxonomy, Vanuatu stands poised to translate policy commitments into culturally resonant and practically effective outcomes. This approach not only advances climate justice for vulnerable island communities but also provides a valuable model for the broader Pacific region as it confronts the escalating challenges of climate-induced loss and damage. Similarly, Fiji's pioneering initiatives illustrate this potential. Fiji has developed national Planned Relocation Guidelines and SOPs, underpinned by a Climate Relocation of Communities Trust Fund, to ensure that adaptation measures uphold community cohesion, respect for vanua, and spiritual well-being [23]. By integrating Indigenous values into policy and creating tools like the CRVAM to holistically assess relocation needs, Fiji is operationalising NELD within its climate finance decisions [43]. These efforts are still evolving, yet they demonstrate how Pacific nations can define solutions on their own terms. With adequate international support, for instance, through an FRLD that prioritises local priorities, such approaches could reshape climate finance to safeguard not just physical assets but also the cultural, spiritual, and social fabrics that define Pacific identities.

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